

ECONOMETRICS I
(Wednesday 10.00-12.50)

Syllabus

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Classnotes: <https://github.com/htastan/Econometrics-I>
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Office Hours: Wednesdays, 14.00-16.00
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COURSE DESCRIPTION & OBJECTIVES

Welcome to Econometrics I! This course is your introduction to the essential tools used to understand and quantify the economic world around us. Have you ever wondered:

- How does an additional year of education actually affect future wages?
- What is the real impact of a new government policy on unemployment rates?
- How do marketing campaigns influence company sales?

Econometrics provides the methods to answer these questions and many more. It's the science of using data to estimate economic relationships, test theories, and evaluate policies. We'll bridge the gap between abstract economic theory and real-world, observable data.

As Arthur Goldberger aptly put it, econometrics is "the social science in which the tools of economic theory, mathematics, and statistical inference are applied to the analysis of economic phenomena." A key challenge we'll tackle is that economists rarely have the luxury of conducting controlled experiments. We can't simply create two identical economies and apply a policy to only one. Therefore, this course focuses on the special statistical methods developed to draw reliable conclusions from the non-experimental data that the real world provides.

By the end of this semester, you will be able to:

- Understand the principles of the classical linear regression model and its assumptions.
- Estimate and interpret simple and multiple regression models using real data.
- Perform hypothesis testing to evaluate the statistical significance of your findings.
- Identify and begin to address common issues in econometric analysis, such as omitted variable bias, heteroskedasticity, and functional form misspecification.
- Gain practical experience in econometric analysis using the professional software R, preparing you for academic research and industry roles.

ECONOMETRICS SOFTWARE: R & RSTUDIO

To bring econometric theory to life, we will use **R**, a powerful, free, and open-source software environment for statistical computing. **R** is the industry standard for statisticians, data scientists, and econometricians, and proficiency in it is a highly valuable skill.

R: Please download and install the base R software from the official repository:
<https://www.r-project.org/>

RStudio: We will use RStudio as our primary interface. It's an integrated development environment (IDE) that makes using R much more intuitive and productive. Download the free RStudio Desktop version here: <https://posit.co/download/rstudio-desktop/>

SUPPLEMENTAL LEARNING: DATACAMP

To support your journey in learning R (and many other software tools), I'm providing you with free access to DataCamp's learning platform. This is a fantastic resource with interactive exercises for learning R programming at your own pace, and it also offers courses in Python, SQL, and other valuable data science skills.

Action Required: You will receive an invitation link to your **std.yildiz.edu.tr** university email. Please use this link and your university email to create your account. While not graded, I will suggest specific DataCamp modules to complete before we tackle certain topics in class. Completing these will make our in-class R sessions much smoother and more effective. Once enrolled, please check the "Assignments" section on the DataCamp platform.

PREREQUISITES

A solid foundation in statistics and mathematics is crucial for success in this course. The formal prerequisites are:

- Statistics I & II
- Mathematics I & II (specifically, college-level algebra and basic calculus)

We will be building directly on concepts like probability distributions, hypothesis testing, and derivatives from day one.

COURSE MATERIALS

Required Text: J.M. Wooldridge (**W**), *Introductory Econometrics: A Modern Approach*, 6th ed., 2016, Cengage Learning.

R applications based on the Wooldridge's text:

- F. Heiss (**H**), *Using R for Introductory Econometrics*, 2016, CreateSpace.
Online version and other materials available at: <http://www.urfie.net/>

Additional Resources:

- Class notes (see the class website)
- J. H. Stock and M.W. Watson (**SW**), *Introduction to Econometrics*, 3rd ed., 2015, Pearson.

EVALUATION

Your final grade will be determined by your performance on two midterm exams and a final exam. The exams will test your understanding of both the theoretical concepts and your ability to interpret econometric output.

Midterms: 60% (two midterms, 30% each); Final: 40%

CLASS SCHEDULE (W: Wooldridge, H: Heiss, SW: Stock and Watson)

Week	Topics	Preparation
Week 1 (Oct. 1)	Introduction to Econometrics: Definition, scope and purpose of econometrics, Types of economic data, Causality and the notion of ceteris paribus, Review of concepts in statistics and mathematics. Lab: Complete Datacamp course: https://app.datacamp.com/learn/courses/free-introduction-to-r	W: ch. 1 SW: ch. 1-2-3 H: ch. 1
Week 2 (Oct. 8)	Simple (Bivariate) Regression Model I: Estimation problem, Ordinary Least Squares (OLS) Estimation, Algebra of OLS Lab: Handling data in R, bivariate regression analysis using R	W: ch. 2 SW: ch. 4 H: ch. 2
Week 3 (Oct. 15)	Simple (Bivariate) Regression Model II: Incorporating nonlinearities, Finite Sample Properties of OLS estimators: Unbiasedness, Efficiency, variance of OLS estimators, t-test, confidence interval Lab: bivariate regression analysis using R	W: ch. 2 SW: ch. 4-5 H: ch.2
Week 4 (Oct. 22)	Multiple Linear Regression Model I: Estimation, Assumptions of the classical model, Gauss-Markov Theorem Lab: multiple regression analysis using R	W: ch. 3 SW: ch. 6 H: ch. 3
Week 5 (Oct. 29)	Multiple Linear Regression Model II: Statistical Inference, Hypothesis Tests, t-test Lab: conducting tests in multiple regression analysis using R	W: ch. 4 SW: ch.7 H: ch. 4
Week 6 (Nov. 5)	Multiple Linear Regression Model III: testing multiple linear restrictions, F-test, Lagrange Multiplier (LM) test Lab: hypothesis tests in regression analysis using R	W: ch. 4 H: ch. 4
Week 7 (Nov. 12)	Asymptotic properties of OLS estimators, Consistency and asymptotic normality Lab: R applications and solutions to selected problems	W: ch. 5 H: ch. 5
Week 8 (Nov. 19)	Midterm I (exact date and time to be announced later)	
Week 9 (Nov. 26)	Functional form in the multiple regression model, quadratic terms in the regression, interaction terms, goodness-of-fit measures, prediction Lab: R applications and exercises	W: ch. 6 H: ch. 6
Week 10 (Dec. 3)	Incorporating qualitative information in the model: Dummy (binary) variables in regression models Lab: R applications and exercises	W: ch. 7 H: ch. 7
Week 11 (Dec. 10)	Binary dependent variables: linear probability model (LPM), Logit model Lab: R applications and solutions to selected problems	W: ch. 7 H: ch. 7
Week 12 (Dec. 17)	Midterm II (exact date and time to be announced later)	
Week 13 (Dec. 24)	Heteroskedasticity (Lab: R applications on heteroskedasticity)	W: ch. 8 H: ch. 8
Week 14 (Dec 31)	Misspecification and data problems	W: ch. 9 H: ch. 9
Week 15 (Jan. 7)	Misspecification and data problems (cont'd) Lab: R applications and solutions to selected problems	W: ch. 9 H: ch. 9
	Final exams	